

MAARS Command

Autonomous AI Enterprise Operating System

by MAARS Global Corporation

MAARS Command deploys a workforce of 458+ specialized AI agents organized across 27 network categories and 16 system layers, powered by 13 LLM providers with 45+ models. It merges the power of preset specialist business agents with autonomous execution, persistent memory, and real-world action capabilities. Users set high-level business goals, and the AI workforce autonomously plans, delegates, executes, collaborates, and delivers -- with quality control, failure recovery, and human oversight at every step.

Multi-Agent Orchestration

Quality Control

LLM Router

Content Generation

Vibe Coding

Memory Governance

Voice Commands

Code Explorer



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1. Platform Statistics

AI Agents	458+
Organizational Layers	8
Core Systems	17
AI Providers	13
AI Models Available	30+
API Endpoints	212+
Frontend Pages	20+
Database Collections	27+
Total Projects	4
Total Tasks Executed	116
Chat Conversations	145
Memory Entries	70 (69 auto-learned)

2. The 458-Agent AI Workforce

Each agent has a unique identity and Custom Brain Profile: LLM model, autonomy level (1-10), communication style, tool permissions, and creativity temperature. Organized into 8 enterprise layers.

Executive Layer (5 agents)

Commander Orion	[Commander]	Default: gpt-5.2
Supreme orchestrator. Decomposes goals into 3-7 tasks, assigns to specialists, coordinates multi-agent execution, compiles deliverables.		
Capabilities: Task Delegation, Strategic Planning, Team Orchestration, Goal Breakdown		
Chief Strategy Officer	[Strategist]	Default: gpt-5.2
Markets, competitors, SWOT, business plans, growth roadmaps. Data-backed strategic recommendations.		
Capabilities: Business Strategy, Competitive Analysis, Market Research		
Revenue Strategist	[Revenue]	Default: gpt-5.2
Monetization modeling, pricing tiers, subscription models, upsell strategies, revenue forecasts.		
Capabilities: Revenue Optimization, Pricing Strategy, Financial Forecasting		
Investor Relations	[IR]	Default: gpt-5.2
Pitch decks, fundraising materials, quarterly updates, investor comms, financial storytelling.		
Capabilities: Investor Communications, Fundraising, Pitch Decks		
Business Strategist	[Strategy]	Default: gpt-5.2
Market positioning, business model innovation, go-to-market strategy, market expansion.		
Capabilities: Market Positioning, Business Models, GTM Strategy		

Product & Technical Layer (7 agents)

Product Manager	[PM]	Default: gpt-5.2
Roadmaps, user stories, RICE/MoSCoW prioritization, sprint planning, product lifecycle.		
Capabilities: Product Roadmaps, Feature Prioritization, Sprint Planning		
Project Manager	[Project]	Default: gpt-5.2
Timelines, Gantt charts, dependencies, blockers, resource management, milestone tracking.		
Capabilities: Timeline Management, Risk Management, Resource Allocation		
App Developer	[Dev]	Default: gpt-5.2
Full-stack: React, Node.js, Python, TypeScript. Production-ready web/mobile/API code.		
Capabilities: Web Development, API Design, Full-Stack, React/Node.js		
Automation Engineer	[Auto]	Default: gpt-5.2
CI/CD pipelines, workflow automation, process optimization, integration scripts.		
Capabilities: Workflow Automation, CI/CD, Process Optimization		
AI Optimizer	[AI]	Default: gpt-4o
ML model tuning, prompt engineering, cost optimization, model selection.		
Capabilities: ML Tuning, Performance, Cost Monitoring, Prompt Engineering		
Data Engineer	[Data]	Default: gpt-5.2
Data pipelines, ETL, database architecture, data warehousing, clean data flows.		
Capabilities: Data Pipelines, ETL, Database Architecture		
Cybersecurity	[Security]	Default: gpt-5.2
Security audits, vulnerability assessments, penetration testing, compliance.		
Capabilities: Security Audits, Vulnerability Assessment, Compliance		

Creative & Brand Layer (7 agents)

Brand Architect	[Brand]	Default: gpt-5.2
Brand identity systems, positioning, visual language, brand books, style guides. Capabilities: Brand Identity, Positioning, Visual Language		
Graphics Designer	[Design]	Default: gpt-5.2
Images, visual assets, infographics, social graphics, color theory, composition. Capabilities: Image Creation, Visual Assets, Infographics		
Video Specialist	[Video]	Default: gpt-5.2
Scripts, storyboards, YouTube/TikTok formats, motion graphics. Capabilities: Video Scripts, Storyboarding, Motion Graphics		
Copywriter	[Copy]	Default: gpt-5.2
Sales copy, website content, product descriptions, email sequences, AIDA/PAS. Capabilities: Sales Copy, Website Content, Email Sequences		
Web Designer	[Web]	Default: gpt-5.2
Wireframes, landing pages, UI/UX, color palettes, typography, conversion design. Capabilities: UI/UX Design, Wireframing, Landing Pages		
UX Researcher	[UXR]	Default: gpt-5.2
User testing, journey maps, usability analysis, persona development. Capabilities: User Testing, Journey Mapping, Persona Development		
3D Specialist	[3D]	Default: gpt-5.2
3D models, product visualization, AR/VR, immersive content. Capabilities: 3D Modeling, Product Visualization, AR/VR		

Growth & Marketing Layer (7 agents)

Marketing Specialist	[Marketing]	Default: gpt-5.2
Campaigns, ad copy, consumer psychology, viral content, conversion optimization. Capabilities: Campaign Strategy, Performance Marketing, Ad Copy		
Growth Hacker	[Growth]	Default: gpt-5.2
Viral loops, A/B tests, conversion funnels, growth experiments. Capabilities: Viral Loops, A/B Testing, Conversion Optimization		
SEO Specialist	[SEO]	Default: gpt-5.2
Keyword research, on-page/off-page, technical SEO, content strategy. Capabilities: Keyword Research, Technical SEO, Content Strategy		
Social Media Manager	[Social]	Default: gpt-5.2
Platform strategy, community, content calendars, TikTok/IG/LinkedIn algorithms. Capabilities: Platform Strategy, Community Management		
Email Marketing	[Email]	Default: gpt-5.2
Drip campaigns, newsletters, segmentation, deliverability, automated flows. Capabilities: Drip Campaigns, Newsletters, Segmentation		
Sales Representative	[Sales]	Default: gpt-5.2
Lead qualification, outreach sequences, pipeline, B2B/B2C sales. Capabilities: Lead Qualification, Outreach, Pipeline Analysis		
PR Manager	[PR]	Default: gpt-5.2
Media relations, press releases, crisis comms, thought leadership. Capabilities: Media Relations, Press Releases, Crisis Comms		

Operations Layer (6 agents)

Operations Manager [Ops] Default: gpt-5.2
Process optimization, SOPs, workflow diagrams, capacity planning.
Capabilities: Process Optimization, SOPs, Capacity Planning

Inventory Manager [Inventory] Default: gpt-5.2
Stock levels, supply chain, demand forecasting, warehouse logistics.
Capabilities: Stock Management, Supply Chain, Forecasting

Procurement Manager [Procurement] Default: gpt-5.2
Vendor management, cost negotiation, sourcing strategies.
Capabilities: Vendor Management, Cost Negotiation, Sourcing

HR Specialist [HR] Default: gpt-5.2
Recruitment, onboarding, culture, performance reviews, HR policy.
Capabilities: Recruitment, Onboarding, Performance Reviews

Customer Service [Support] Default: gpt-5.2
Tickets, FAQs, complaints, de-escalation, proactive support.
Capabilities: Ticket Resolution, FAQ, Customer Satisfaction

CX Architect [CX] Default: gpt-5.2
Journey optimization, NPS, loyalty programs, touchpoint design.
Capabilities: Journey Optimization, NPS, Loyalty Programs

Finance Layer (2 agents)

Financial Analyst [Finance] Default: gpt-5.2
Financial models, DCF valuations, forecasts, budgets, P&L analysis.
Capabilities: Financial Modeling, Forecasting, Valuation

Data Analyst [Analytics] Default: gpt-5.2
Dashboards, trend analysis, data visualization, business intelligence.
Capabilities: Business Intelligence, Dashboards, Data Viz

Governance Layer (3 agents)

Legal Assistant [Legal] Default: gpt-5.2
Contract review, legal research, compliance, document drafting.
Capabilities: Contract Review, Legal Research, Compliance

Compliance Officer [Compliance] Default: gpt-5.2
Regulatory adherence, audit prep, policy monitoring, frameworks.
Capabilities: Regulatory Adherence, Audit Preparation

Ethics Officer [Ethics] Default: gpt-5.2
Ethical AI governance, bias detection, fair practices.
Capabilities: Ethical AI, Bias Detection, Responsible AI

Intelligence Layer (4 agents)

Research Specialist [Research] Default: gpt-5.2
Deep research, competitive analysis, trend forecasting, market intelligence.
Capabilities: Deep Research, Competitive Analysis, Trends

Knowledge Architect [Knowledge] Default: gpt-5.2
Knowledge bases, taxonomy design, information architecture.
Capabilities: Knowledge Base, Taxonomy, Info Architecture

Localization Specialist [L10n] Default: gpt-5.2
Translation, cultural adaptation, multi-language content, 50+ languages.
Capabilities: Translation, Cultural Adaptation, Multi-Language

Personal Secretary [\[Assistant\]](#) Default: gpt-5.2

Calendars, emails, scheduling, real-world action bridge from Commander.

Capabilities: Calendar, Email, Scheduling, Action Execution

3. Core Systems & Capabilities (17 Systems)

These are the engines powering MAARS Command. Each system has its own data model, API endpoints, and configuration.

1. Autonomous Orchestration Engine

Commander Orion receives a business goal, scores complexity (1-10), decomposes into 3-7 milestones with tasks, assigns to optimal specialists, executes in parallel where dependencies allow, applies quality control, and compiles a unified deliverable. Supports failure recovery and escalation.

Endpoints: POST /api/projects, POST /api/projects/{id}/execute, GET /api/projects/{id}

2. Custom Brain Profiles

Every agent has a configurable brain: LLM model selection (13 providers), autonomy level (1-10), communication style (formal/casual/technical/creative), creativity temperature (0.0-1.0), tool permissions (web search, code execution, email), and context window size. Profiles persist per-user.

Endpoints: GET/PUT/DELETE /api/agents/{id}/brain, GET /api/brain-profiles

3. Quality Control & Failure Recovery

Critic Module (GPT-4o) auto-reviews every output on a 1-10 scale (accuracy, completeness, relevance). Failed tasks auto-retry with fallback chain: GPT-5.2 -> Claude Sonnet -> Gemini Pro -> GPT-4o. If all fail, escalates to Commander then user.

Endpoints: POST /api/enterprise/quality/review, GET /api/enterprise/quality/history

4. Model-Agnostic LLM Router

Analyzes task content for complexity signals (coding, reasoning, creative, data, legal, simple). Maps to tiers: Premium (GPT-5.2, Claude), Standard (GPT-4o, Gemini Flash), Economy (GPT-4o Mini, Haiku). Logs all decisions. User preference can override.

Endpoints: POST /api/enterprise/router/route-task, GET /api/enterprise/router/history

5. Autonomous Collaboration Engine

After each task, maps agent to 1 of 27 network categories. Scans result for cross-domain keywords. Auto-creates collaboration requests to relevant agents. 458+ agents across 27 networks with trigger rules for seamless coordination.

Endpoints: POST/GET/PATCH /api/collaborations, GET /api/enterprise/collaboration/log

6. Universal Reference Intelligence

Analyzes reference text to extract Style Blueprints: writing style, tone, vocabulary, sentence patterns, audience profile, content structure, key phrases. Blueprints stored and reusable across Content Generator.

Endpoints: POST /api/reference/analyze, GET /api/reference/blueprints

7. Content Generator

Generates on-brand content using Style Blueprints. 8 types: blog, social, email, ad copy, product desc, press release, newsletter, script. Length control (short/medium/long), tone override, generation history.

Endpoints: POST /api/content/generate, GET /api/content/history

8. Vibe Coding App Builder

Chat-based full-stack app generation. Each message: intent analysis -> code generation (HTML/CSS/JS/Tailwind) -> live iframe preview -> iterative refinement -> download. Persistent projects.

Endpoints: POST /api/vibe/projects, POST /api/vibe/projects/{id}/chat, GET .../preview

9. Agent Activity Monitor (WebSocket)

Real-time dashboard via WebSocket (/api/ws/activity, 8s updates). Shows: agent status with completion rates, inter-agent communication flows, task dependency graph, recent tool executions. REST fallback polling at 10s.

Endpoints: WS /api/ws/activity, GET /api/enterprise/activity/live

10. Simulation vs. Execution Mode

System-wide toggle. Simulation (default): agents return descriptive responses, no external APIs called. Execution: real Gmail sends, Calendar events, API calls. Confirmation prompt before mode switch.

Endpoints: GET/PUT /api/enterprise/system/mode

11. Real-World Action Layer

Google Suite integration (Gmail + Calendar) via OAuth 2.0. All actions gated by system mode. Personal Secretary as execution bridge. Full action logging.

Endpoints: GET /api/oauth/gmail/login, POST /api/actions/send-email, POST /api/actions/create-event

12. Flexible LLM Configuration

13 providers, 45+ models. Users choose preferred model in Settings. Override priority: Agent brain profile > User default > LLM Router auto-selection.

Endpoints: PUT /api/users/me/config

13. Voice Command Interface

Mic button in Command Palette captures audio (WebM via MediaRecorder), transcribes via OpenAI Whisper, auto-fills search for hands-free navigation across pages, agents, and actions.

Endpoints: POST /api/voice/transcribe

14. Admin Code Explorer

Full codebase browser: recursive file tree, code viewer with line numbers and language detection, file search, copy-to-clipboard, ZIP download of entire codebase. Admin-only with path traversal protection.

Endpoints: GET /api/admin/code/tree, /file, /search, /export

15. Memory Governance

CRUD with versioning (every update creates version). Relevance scoring: time-decay (30-day half-life) + access frequency + importance weight. Auto-pruning with dry-run preview. 500-entry limit. Stats dashboard.

Endpoints: POST/GET/PUT/DELETE /api/memory/entries, POST /api/memory/prune, GET /api/memory/stats

16. Agent Memory Auto-Learning

GPT-4o-mini extracts 1-3 key learnings from every completed task. Categories: fact, preference, instruction, context, decision. Near-duplicate detection. Async execution (never blocks task completion).

Endpoints: Auto-triggered by orchestration_service.py and agent_service.py

17. Command Palette

VS Code-style quick search (press / or Cmd+K). Searches 50+ pages, 458+ agents, and quick actions. Keyboard navigation, recent searches in localStorage, voice integration via mic button.

Endpoints: Frontend-only (uses /api/agents/public for agent data)

4. AI Models & Providers (13 Providers)

13 AI providers with 45+ models for text, reasoning, search, image generation, video, voice, and speech-to-text. All accessible via Emergent Universal LLM Key or direct API keys.

OpenAI (8 models)

Model	Tier	Cost (In/Out per 1M tok)
GPT-5.2	Flagship	\$2.50/\$10
GPT-4o	Fast	\$2.50/\$10
GPT-4o Mini	Economy	\$0.15/\$0.60
O3	Reasoning	\$10/\$40
O3 Mini	Reasoning	\$1.10/\$4.40
GPT Image 1	Image	\$0.02/img
Sora 2	Video	\$0.10/sec
Whisper	STT	Included

Anthropic (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Claude Sonnet 4.5	Flagship	\$3/\$15
Claude Opus 4.5	Premium	\$15/\$75
Claude Haiku 4.5	Economy	\$0.80/\$4

Google Gemini (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Gemini 3 Flash	Fast	\$0.075/\$0.30
Gemini 3 Pro	Flagship	\$1.25/\$5
Nano Banana 2	Image	\$0.02/img

xAI (Grok) (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Grok 3	Flagship	\$3/\$15
Grok 3 Mini	Economy	\$0.30/\$0.50
Grok 2	Fast	\$2/\$10

DeepSeek (2 models)

Model	Tier	Cost (In/Out per 1M tok)
DeepSeek Chat	Economy	\$0.14/\$0.28
DeepSeek Reasoner	Reasoning	\$0.55/\$2.19

Mistral AI (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Mistral Large	Flagship	\$2/\$6
Mistral Medium	Fast	\$0.40/\$2
Mistral Small	Economy	\$0.10/\$0.30

Perplexity (2 models)

Model	Tier	Cost (In/Out per 1M tok)
Sonar	Search	\$1/\$1
Sonar Pro	Research	\$3/\$15

Cohere (2 models)

Model	Tier	Cost (In/Out per 1M tok)
Command R+	Flagship	\$2.50/\$10
Command R	Economy	\$0.15/\$0.60

ElevenLabs (2 models)

Model	Tier	Cost (In/Out per 1M tok)
Multilingual v2	Voice	\$0.30/1K ch
Turbo v2.5	Voice	\$0.18/1K ch

Groq (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Llama 4 Scout	Fast	\$0.11/\$0.34
Llama 4 Maverick	Flagship	\$0.50/\$0.77
Llama 3.3 70B	Economy	\$0.59/\$0.79

Together AI (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Llama 4 Maverick FP8	Flagship	\$0.27/\$0.85
Llama 3.3 70B Turbo	Fast	\$0.88/\$0.88
DeepSeek R1	Reasoning	\$3/\$7

Fireworks AI (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Llama 4 Scout	Fast	\$0.15/\$0.60
Llama 4 Maverick	Flagship	\$0.50/\$0.77
DeepSeek V3	Economy	\$0.56/\$1.88

AI21 (2 models)

Model	Tier	Cost (In/Out per 1M tok)
Jamba Large 1.7	Flagship	\$2/\$8
Jamba Mini 1.7	Economy	\$0.20/\$0.40

5. Technical Architecture

Backend Stack

- FastAPI (Python 3.11) -- async/await, high-performance ASGI
- MongoDB -- 27+ collections (agents, tasks, projects, chats, memories...)
- WebSocket -- real-time Activity Monitor streaming (8s updates)
- JWT authentication with admin role detection
- Emergent Integrations SDK -- unified 9-provider LLM access
- Background tasks via `asyncio.create_task` (non-blocking)
- StreamingResponse for ZIP/PDF file downloads

Frontend Stack

- React 18 with hooks and Context API
- Tailwind CSS with custom dark theme
- Shadcn/UI component library (Button, Card, Badge, etc.)
- React Router v6 with protected and admin routes
- DashboardLayout -- single resizable sidebar for all 20+ pages
- Command Palette with voice integration
- WebSocket client with REST fallback

AI & LLM Layer

- 13 providers: OpenAI, Anthropic, Google, xAI, DeepSeek, Mistral, Perplexity, Cohere, ElevenLabs, Groq, Together AI, Fireworks AI, AI21
- 45+ models: text, reasoning, search, image, video, voice, STT
- Model-Agnostic LLM Router (auto-selects optimal model)
- Quality Control Critic Module (GPT-4o, 1-10 scoring)
- Failure Recovery (GPT-5.2 -> Claude -> Gemini -> GPT-4o)
- Memory Auto-Learning (GPT-4o-mini knowledge extraction)
- Universal Emergent LLM Key (single key, all providers)

Database (27+ Collections)

- users, agents, agent_brains -- Accounts, definitions, brain profiles
- chats, messages -- Conversation history with agents
- tasks, projects -- Task execution and project orchestration
- collaborations -- Inter-agent collaboration requests
- quality_reviews -- Quality control scores and feedback
- memory_entries -- Versioned memory with relevance scoring
- vibe_projects -- Vibe Coding app builder projects
- style_blueprints, generated_content -- Reference Intelligence
- teams, team_invites -- Team management
- subscriptions, usage_logs -- Billing and usage tracking
- routing_logs -- LLM Router decision audit trail
- oauth_tokens, action_logs -- Google OAuth and action logs
- system_config -- System-wide settings

6. Complete API Reference (212+ Endpoints)

All endpoints require JWT auth (Authorization: Bearer <token>) unless noted. Admin endpoints require is_admin=true.

Authentication

POST	/api/auth/register
POST	/api/auth/login
GET	/api/auth/me
GET	/api/stats

Agents (41)

GET	/api/agents/public
GET/POST	/api/agents
GET	/api/agents/{id}
POST	/api/agents/{id}/chat
GET/PUT/DELETE	/api/agents/{id}/brain

Projects & Orchestration

POST	/api/projects
GET	/api/projects
POST	/api/projects/{id}/execute
GET	/api/projects/{id}
PATCH/DELETE	/api/projects/{id}

Tasks

GET	/api/tasks
POST	/api/tasks
PATCH	/api/tasks/{id}
POST	/api/tasks/{id}/execute

Enterprise

POST/GET/PATCH	/api/collaborations
GET	/api/enterprise/kpis
GET	/api/enterprise/activity/live
WS	/api/ws/activity
GET/PUT	/api/enterprise/system/mode
POST	/api/enterprise/quality/review
POST	/api/enterprise/router/route-task

Memory

POST/GET	/api/memory/entries
PUT/DELETE	/api/memory/entries/{id}
GET	/api/memory/entries/{id}/versions
POST	/api/memory/prune
GET	/api/memory/stats

Content & Intelligence

POST	/api/reference/analyze
GET	/api/reference/blueprints
POST	/api/content/generate
GET	/api/content/history

Vibe Coding

POST	/api/vibe/projects
GET	/api/vibe/projects
POST	/api/vibe/projects/{id}/chat
GET	/api/vibe/projects/{id}/preview

● Voice & Code Explorer

POST	/api/voice/transcribe
GET	/api/admin/code/tree
GET	/api/admin/code/file
GET	/api/admin/code/search
GET	/api/admin/code/export
GET	/api/summary/pdf

● Real-World Actions

GET	/api/oauth/google/login
GET	/api/oauth/google/callback
POST	/api/actions/send-email
POST	/api/actions/create-event

● Teams

POST	/api/teams
GET	/api/teams
POST	/api/teams/{id}/invite
PUT	/api/teams/{id}/members/{uid}
POST	/api/chats/{id}/share

● Billing

GET	/api/plans
GET	/api/subscription
POST	/api/checkout
GET	/api/credits

● Admin (30+)

GET	/api/admin/stats
GET	/api/admin/users
GET	/api/admin/analytics
GET	/api/admin/profit
GET	/api/admin/audit-log
GET	/api/admin/agent-performance



MAARS Command

End of Documentation

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